

Hands on Assignment 4

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GitHub link -

<https://github.com/ShreyaRai20/HandsOnAssignment4UseUnsupervisedDeepLearningAlgorithmtDetectFraudwithPyOD.git>

Screenshot-

```
(.venv) shreyarai@Mac HandsOnAssignment4UseUnsupervisedDeepLearningAlgorithmtDetectFraudwithPyOD % python fraud_detection_autoencoder.py --data creditcard.csv --output-dir output
2026-07-02 11:11:41,708 | INFO | Loading real dataset from creditcard.csv
2026-07-02 11:11:42,413 | INFO | Dataset shape: (284807, 31)
2026-07-02 11:11:42,413 | INFO | Fraud cases: 492 (0.1727% of data)
2026-07-02 11:11:42,413 | INFO | Preprocessing data: scaling Amount/Time and splitting train/test
2026-07-02 11:11:42,556 | INFO | Train size: 199364 (fraud=344) | Test size: 85443 (fraud=148)
2026-07-02 11:11:42,556 | INFO | Building AutoEncoder with hidden layers: [30, 16, 16, 30]
2026-07-02 11:11:42,559 | INFO | Training AutoEncoder ...
Training: 100% | 30/30 [02:56<00:00, 5.88s/it]
2026-07-02 11:14:41,637 | INFO | Training complete.
2026-07-02 11:14:41,637 | INFO | Scoring test set and computing metrics ...
2026-07-02 11:14:42,987 | INFO | ROC-AUC: 0.9481
2026-07-02 11:14:42,987 | INFO | Average Precision (PR-AUC): 0.0584

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FRAUD DETECTION - AUTOENCODER EVALUATION RESULTS
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ROC-AUC Score: 0.9481
Average Precision (PR-AUC): 0.0584

Confusion Matrix (rows=actual, cols=predicted):
[[85179 116]
 [ 141 7]]

Classification Report:
      precision    recall  f1-score   support

   Normal         1.00      1.00      1.00     85295
    Fraud         0.06      0.05      0.05       148

 accuracy          0.53      0.52      0.53     85443
 macro avg          0.53      0.52      0.53     85443
weighted avg          1.00      1.00      1.00     85443

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2026-07-02 11:14:43,723 | INFO | Saved plots to: output/roc_curve.png and output/reconstruction_error_hist.png
```